Project Design Document

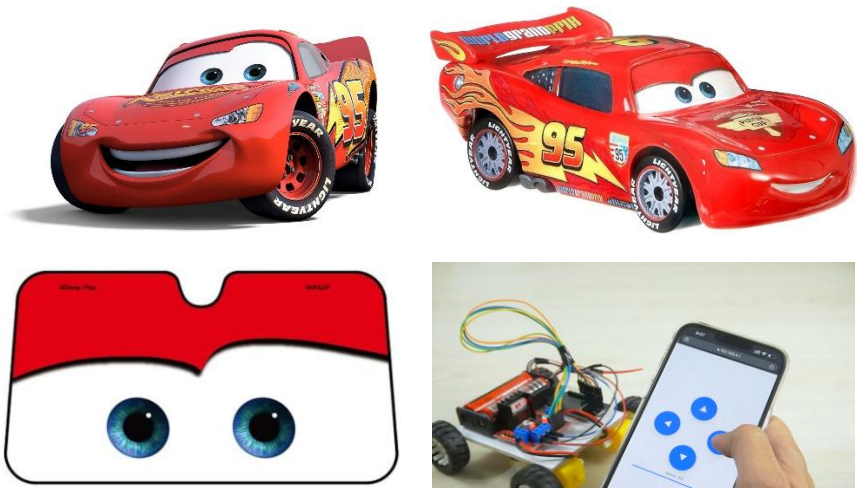
Design Intent

Experience:
Create a playful, responsive toy car that goes beyond basic control by using light and sound to enhance interaction. The synchronized LED indicators and adaptive buzzer feedback aim to create an engaging and intuitive experience, while also building a sense of nostalgia inspired by character-based vehicles.

Target Users:
Kids and fans of Cars, or anyone who enjoys interactive, app-controlled toys.

Inspiration & References

https://youtu.be/Tta4xhmYSwI?si=WCV5FQTvNgDemQX9



- Inspired by Cars for character-like vehicles with expressive eyes.

- Existing RC cars and DIY ESP32 robot cars for movement and control systems.

Learnings:

- We learned how to use a mobile app as a remote control instead of building a physical controller, making the system simpler and more flexible.

- We understood how basic electronics like ESP32 can be used to build interactive connected toys.
- We explored how light and sound can be used to communicate system feedback (turning, reversing, proximity) to the user in an intuitive way.

User Journey

1. User opens the mobile app and connects to the car via WiFi.
2. User presses directional buttons to move the car (forward, backward, left, right).
3. System responds instantly with motion using 4 DC motors controlled by 2 motor drivers.
4. When the user presses **left or right**, the car turns and the LED indicators automatically blink in sync with the direction (no separate indicator buttons).
5. When the user presses **reverse**, the car moves backward and:
    1. The buzzer starts emitting a reverse warning sound
    2. LEDs blink at **0.5-second intervals**
6. As the car approaches an obstacle (detected using the ultrasonic sensor from the rear):
    1. **15 cm:** Slow beeping pattern
    2. **10–15 cm:** Medium speed beeping
    3. **5–10 cm:** Faster beeping
    4. **< 5 cm:** Continuous beep
7. The system continuously provides feedback through motion, light, and sound, making the interaction intuitive, safe, and engaging.

Success Criteria

1. Does the car move correctly when a button is pressed?
2. Do the LEDs and buzzer respond at the right time?
3. Is the system easy and intuitive to use?

Functional

1. Car responds to movement commands (forward, backward, left, right) with minimal delay
2. LED indicators activate correctly and synchronize with left/right turning
3. During reverse, LEDs blink at consistent 0.5-second intervals
4. Buzzer activates during reverse and changes frequency based on distance
5. Ultrasonic sensor reliably detects obstacles and updates buzzer response in real time

User Experience

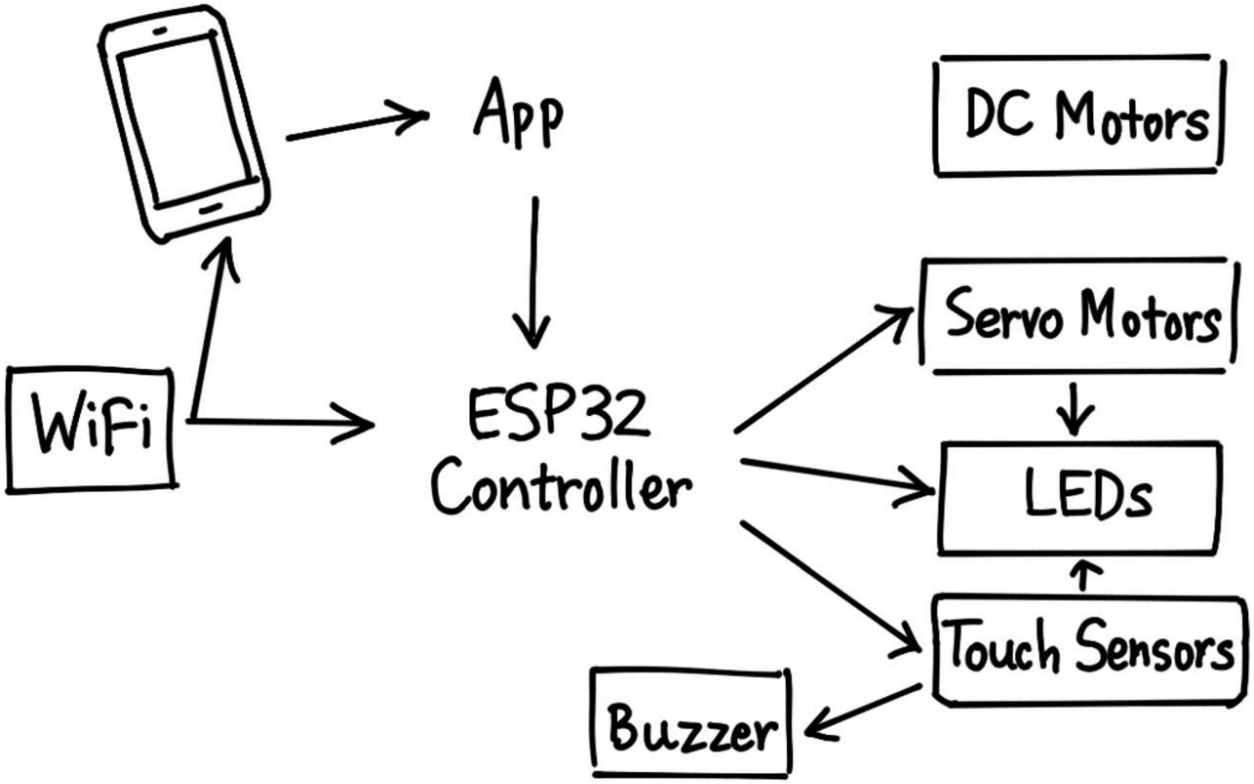
1. Controls are simple, intuitive, and easy to use through the mobile app
2. Clear and immediate feedback through light (LEDs) and sound (buzzer)
3. Reverse assist feature (adaptive buzzer) enhances awareness and usability
4. Car feels responsive and engaging despite removal of character animation
5. System is durable, stable, and safe for repeated use

System Overview

The system is a WiFi-controlled smart toy car built using an ESP32 as the central controller. A mobile app sends commands over WiFi, which the ESP32 processes to control different components. Four DC motors, managed through two motor drivers, handle movement of the car. LED lights are used as indicators and are synchronized with turning and reversing actions. An ultrasonic sensor placed at the rear detects obstacles while reversing, and a buzzer provides adaptive audio feedback based on distance to improve safety.

The system integrates motion, light, and sound to create an interactive and responsive toy experience inspired by Cars.

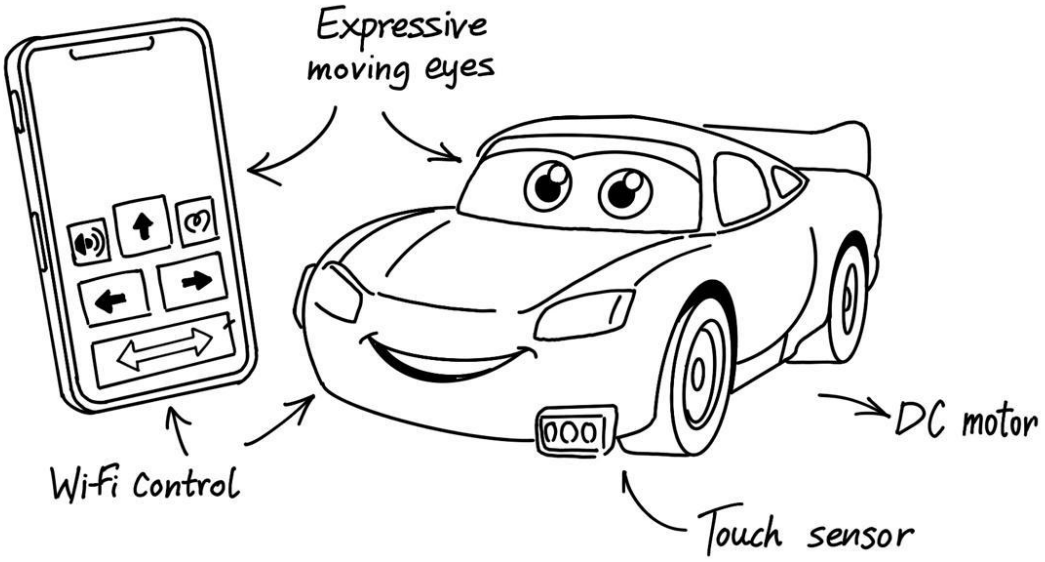
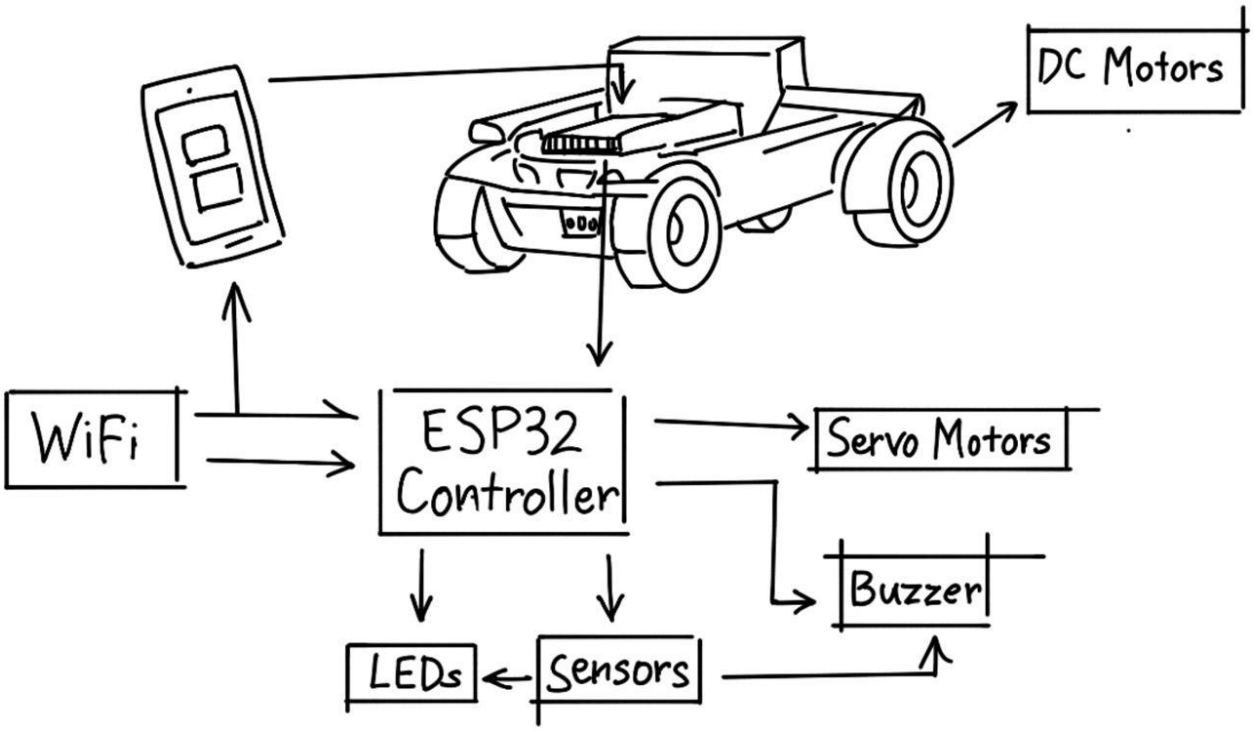
High Level Block Diagram



Input-Output Mapping

Input	Process	Output
1. App button (forward/back/etc.) 2. Indicator button 3. UltraSonic sensor (front/back)	1. Signal mapped to motor driver pins 2. Trigger LED control logic 3. Detect HIGH signal (collision)	1. Car moves in direction 2. Headlights flash 3. Buzzer sound + possible stop

Sketches and Concepts



Final System Architecture